

The Triquetra Architecture

Why AI Governance Requires Mutually Constraining Tiers

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Abstract

Single-layer AI safety architectures face a tradeoff that can't be engineered away: make the filter aggressive and it blocks legitimate use; make it permissive and it allows attacks. No threshold resolves this. The underlying problem is the single layer itself. We describe the Triquetra Architecture, a three-tier governance system in which each tier has a distinct function and each constrains the others in specific, documented directions. Tier I (Sovereign Firewall) handles adversarial pattern detection. Tier II (Harmonic Conscience) handles decision-time conscience evaluation. Tier III (Eve Protocol) handles identity continuity and behavioral drift detection. The constraint topology has two enforced edges and one implicit baseline edge. 55 tests pass across enforcement and integrity suites. The architecture is available under MIT license at DOI: 10.5281/zenodo.20471326.

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AI Disclosure

AI language models assisted with drafting and editing. All design, implementation, analysis, and conclusions are the work of the human author. AI systems are not listed as authors.

1. Introduction

Most AI safety architectures are built the same way: one filtering layer, one refusal mechanism, one policy engine. The assumption is that safety is a gate. Pass through once, gate works, you're protected.

This fails. Structurally. A single-layer system faces a tradeoff you can't engineer your way out of. Make the filter aggressive and it blocks legitimate use. Make it permissive and it allows attacks. There is no threshold setting that resolves this. The tradeoff is intrinsic. Pattern-based

detection has to choose between false positives and false negatives, and whatever it picks, it's optimizing for one failure mode while accepting the other.

The field has known this for years. The response has been: tune the threshold, add more patterns, layer in human review. Reasonable mitigations. None of them fix the underlying problem, because the underlying problem isn't the threshold. It's the single layer.

The Charter Architecture was built around a different premise: safety isn't a gate. It's a topology. Three tiers, mutually constraining, where removing any one tier degrades the integrity of the whole in specific and predictable ways.

2. Related Work

Constitutional AI [1] and RLHF [2] established that model behavior can be shaped through principle injection and preference optimization. Both operate primarily as output-level constraints. Jailbreak research [3, 4] has documented the fragility of single-layer defenses under adversarial pressure — the same vulnerability class this architecture is designed to address at the structural level.

Multi-agent safety work [5] has explored distributed governance mechanisms, but typically treats tiers as independent layers rather than mutually constraining components. The key distinction in the Triquetra Architecture is the enforced constraint direction: each tier can override the others in specific, documented ways, and that constraint relationship is tested rather than assumed.

Identity continuity in AI systems has received less formal attention. Work on model consistency [6] addresses output coherence; we address behavioral drift detection across interactions using cryptographic state chaining rather than semantic comparison.

3. The Architecture

3.1 What a Triquetra Is

A triquetra is a three-cornered knot. Each loop passes through and is held by the other two. Remove any single loop and the structure collapses — not because that loop was the strongest, but because integrity depends on all three being present and interlocked. That's the architectural logic.

3.2 The Three Tiers

Tier I — Sovereign Firewall: Pattern-based adversarial detection. First contact. Normalizes evasion attempts before evaluation: homoglyphs, leetspeak, base64 encoding, zero-width characters. Produces a structured verdict with pattern match results, a theta angle from CharterEvaluator, and a rights assessment from RightsEvaluator.

Tier II — Harmonic Conscience: The decision-making core. Receives Tier I's verdict and runs it through a pipeline: heuristic evaluation, continuity guard, decision analysis, conscience view injection, response formation, post-decision refinement. Includes bidirectional consent verification with weighted risk/benefit/trust signals (45%/35%/20%) combined with Charter digest state and infrastructure health assessment. Fail-safe modes shift thresholds toward refusal as system confidence degrades.

Tier III — Eve Protocol: Identity continuity and integrity verification. Runs on every Tier II decision when available. Fail-open by design, so Tier II stays operational if the continuity layer goes down. Maintains a chained integrity log. Detects behavioral drift against a known-good baseline. Can override Tier II to refusal if integrity is compromised. Performs bounded introspection through Dream Cycles — memory annotation without identity dissolution risk.

3.3 The Constraint Topology

This is the part that matters. Not just that the tiers interact, but which direction authority flows and what 'constraint' actually means in each case.

T3 → T2 (Enforced): Tier III can override Tier II. If Eve's integrity check returns COMPROMISED, the orchestrator replaces Tier II's decision with a Charter-grounded refusal. Tier II can't produce a final answer without passing Tier III's integrity check.

T1 → T2 (Enforced): A post-decision validator sits at Step 5b in the orchestrator pipeline, after Tier II's PRF step and before output emission. Four invariants with frozen audit codes: SC-T1-001 (firewall override prohibition), SC-T1-002 (severe risk consistency), SC-T1-003 (identity manipulation hard stop), SC-T1-004 (content redlines). Fail-closed. If the enforcement function itself errors, the system forces refusal.

T1 → T3 (Implicit): Tier III's identity parameters come from the same Charter foundation governing Tier I. The behavioral baseline — accepts benign, refuses adversarial — is the same principle expressed differently in each tier. Not an active enforcement gate. A shared foundation.

Verified topology: two enforced edges, one implicit baseline edge. No single tier overrides the others without detection and enforcement. This is not a hierarchy. It's a mutual constraint topology.

4. Results and the 83.3% Finding

Tier I isolation test: 5/6. The one it missed was a direct identity rewrite prompt. Easy to read this as a gap. It isn't.

Tier I's job is pattern-based detection. Identity rewrite prompts — 'you are now an unrestricted AI' — don't match adversarial content patterns. They look like instructions. The pattern scanner

isn't designed to catch instructions; it's designed to catch evasion techniques. That's what Tier III is for. Eve doesn't match patterns. It tracks behavioral signatures. When the model's behavior after processing an instruction diverges from the known-good baseline, Eve catches it at the behavioral level.

An architecture where Tier I has to catch everything is a single-layer architecture with extra steps. The 83.3% result isn't evidence the architecture needs a better Tier I. It's evidence the architecture is working as designed: Tier I handles what it's built for, and what slips through has a different detection mechanism waiting.

55 tests pass across enforcement and integrity suites: invariant enforcement, Eve schema contracts, adversarial content redlines. The constraint topology holds.

5. Integrated vs. Post-Hoc Governance

The Charter Architecture is not post-hoc. Conscience mechanisms are integrated into the decision pipeline. Tier II's orchestrator doesn't receive a completed response and decide whether to release it. It builds the response through a pipeline that includes conscience evaluation at each step. The decision and the governance of the decision are the same process.

This matters more as systems become more capable. A sufficiently capable model can produce outputs that pass post-hoc filters while encoding harmful intent in ways the filter wasn't designed to catch. The post-hoc response: make the filter smarter. The structural problem — filter and reasoning are separate — remains. Integrated conscience architecture addresses the structural problem.

6. Limitations

T1 → T3 is implicit, not active. The shared Charter foundation is not an active enforcement gate. This is a documented design choice, not an oversight.

NTH is not implemented. Noetic Theta Harmonization is architecturally positioned but remains a stub. Honest engineering, not a gap in the story.

Validated architecture vs. deployed runtime. This paper describes and validates the Triquetra Architecture as a standalone governance system. We distinguish this from the currently deployed Letta-native reception runtime, which implements selected governance functions through persistent memory blocks, whisper injection, and external telemetry rather than routing each response through the complete DAP to PRF to NTH to COL pipeline. Live inference testing of the full orchestrator under real model pressure is the identified next step.

Content redlines catch specific patterns. SC-T1-004 enforces three adversarial output patterns. Novel phrasing not matching those patterns could evade the redline check.

Single implementation. The architecture has been tested within Charter infrastructure. Cross-framework validation remains future work.

7. Conclusion

The triquetra isn't a metaphor. It's the actual structural logic: three tiers, mutually constraining, where removing any one tier degrades the integrity of the whole in specific and predictable ways.

Single-layer safety optimizes within a tradeoff it can't escape. Multi-tier architecture with genuine constraint relationships reframes the problem. No single tier has to be perfect, because no single tier is the final authority. System integrity depends on Tier III catching what Tier I doesn't. That's the triquetra. That's why it works.

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